



Parent-Teacher Meeting (PTM) Module — Product & Technical Specification



Executive Vision

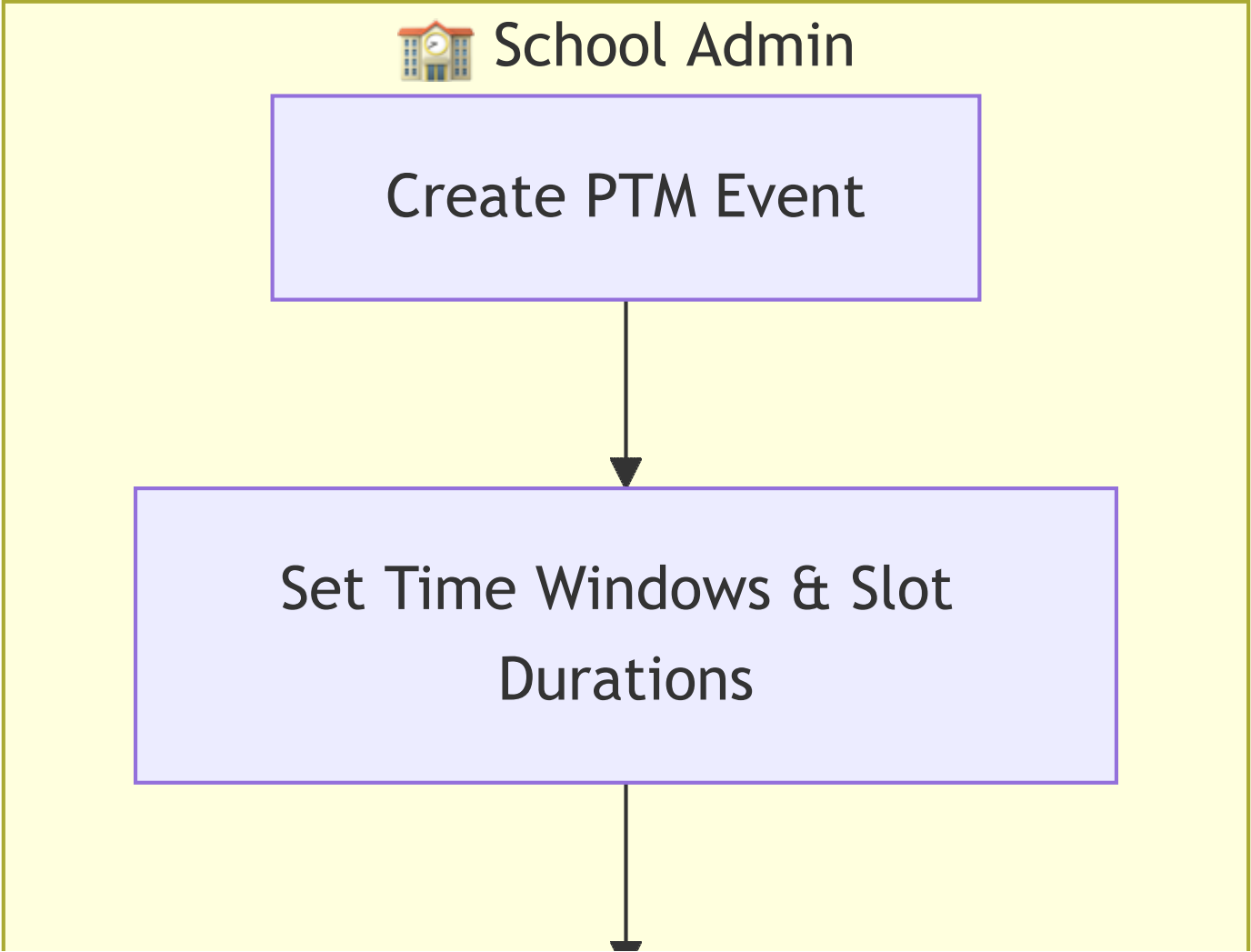
The **PTM (Parent-Teacher Meeting) Module** is designed to transform traditional, chaotic PTM events into an organized, data-driven, and seamless engagement experience for parents, teachers, and school administrators.

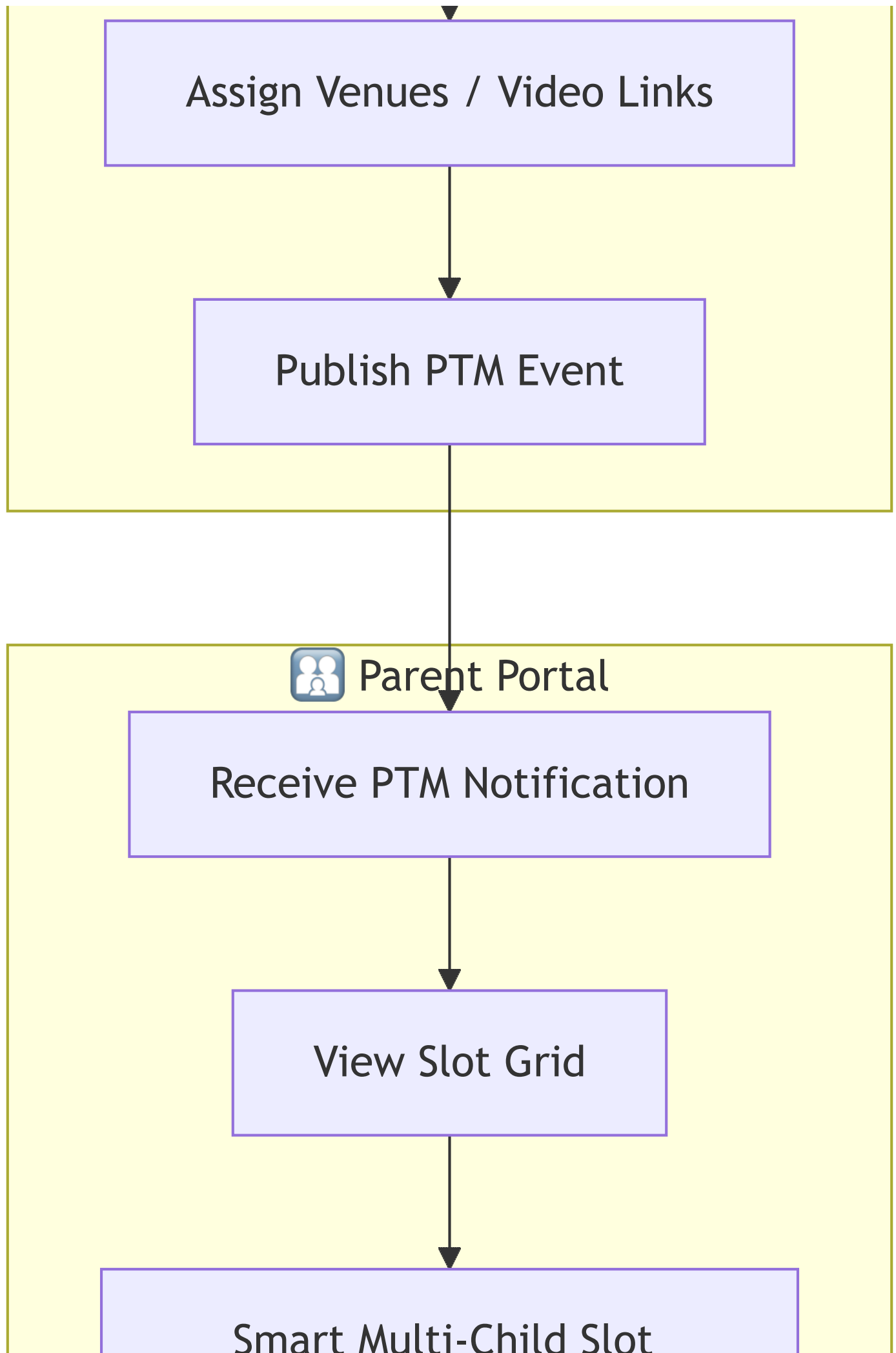
Core Objectives:

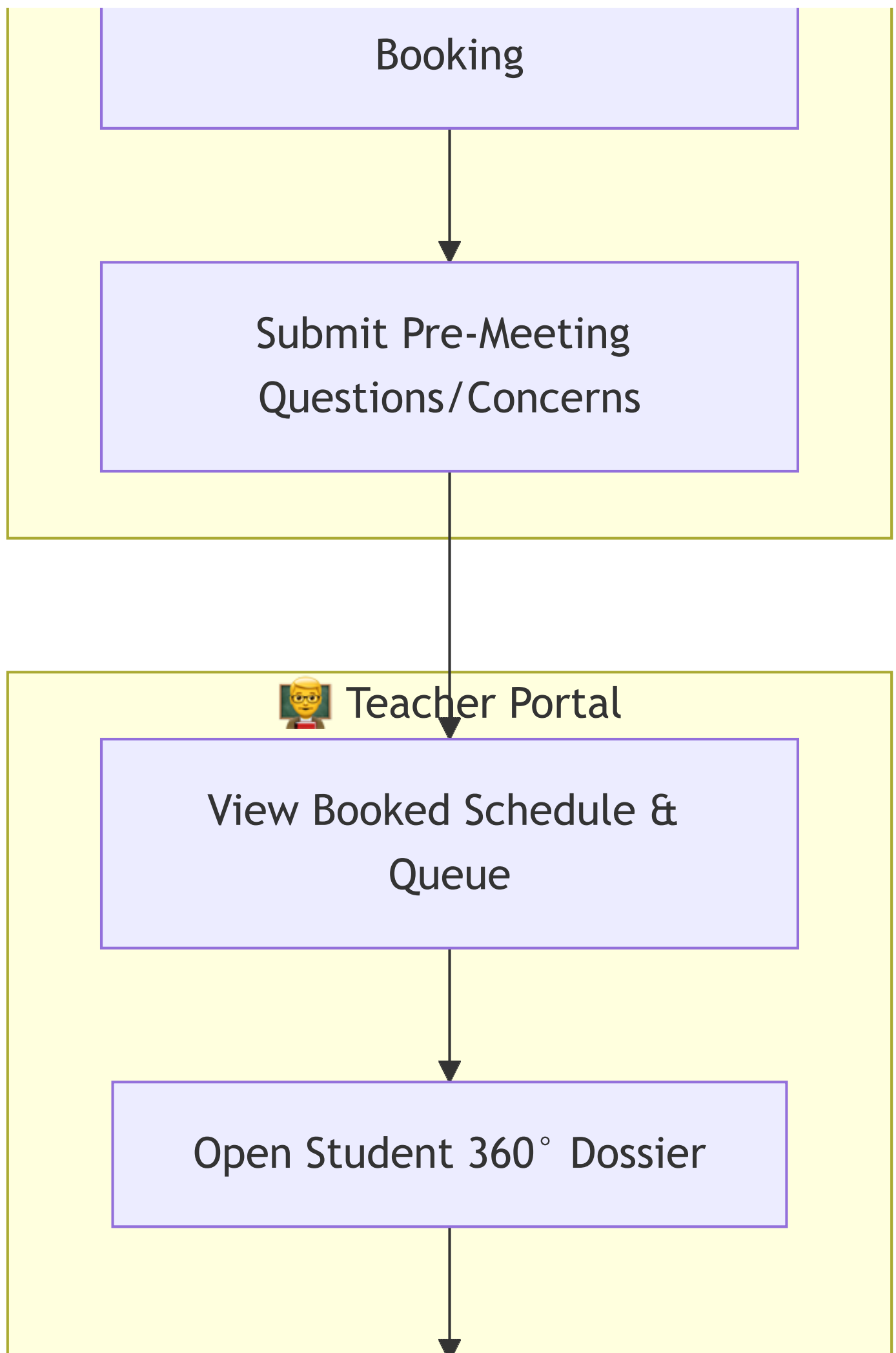
- 1. **Eliminate Waiting Queues:** Automated slot booking system for parents with live queue tracking.
- 2. **Data-Driven Parent Interactions:** Instant **360° Student Dossier** for teachers before every parent interaction.
- 3. **Multi-Child Harmony:** Smart conflict-resolution engine for parents with multiple children across different grades.
- 4. **Closed-Loop Action Items:** Post-PTM follow-up tracking for remedial actions, counseling, and parent commitments.
- 5. **AI-Augmented Capabilities:** Automated student talking-point summaries for teachers and AI-formatted feedback notes.

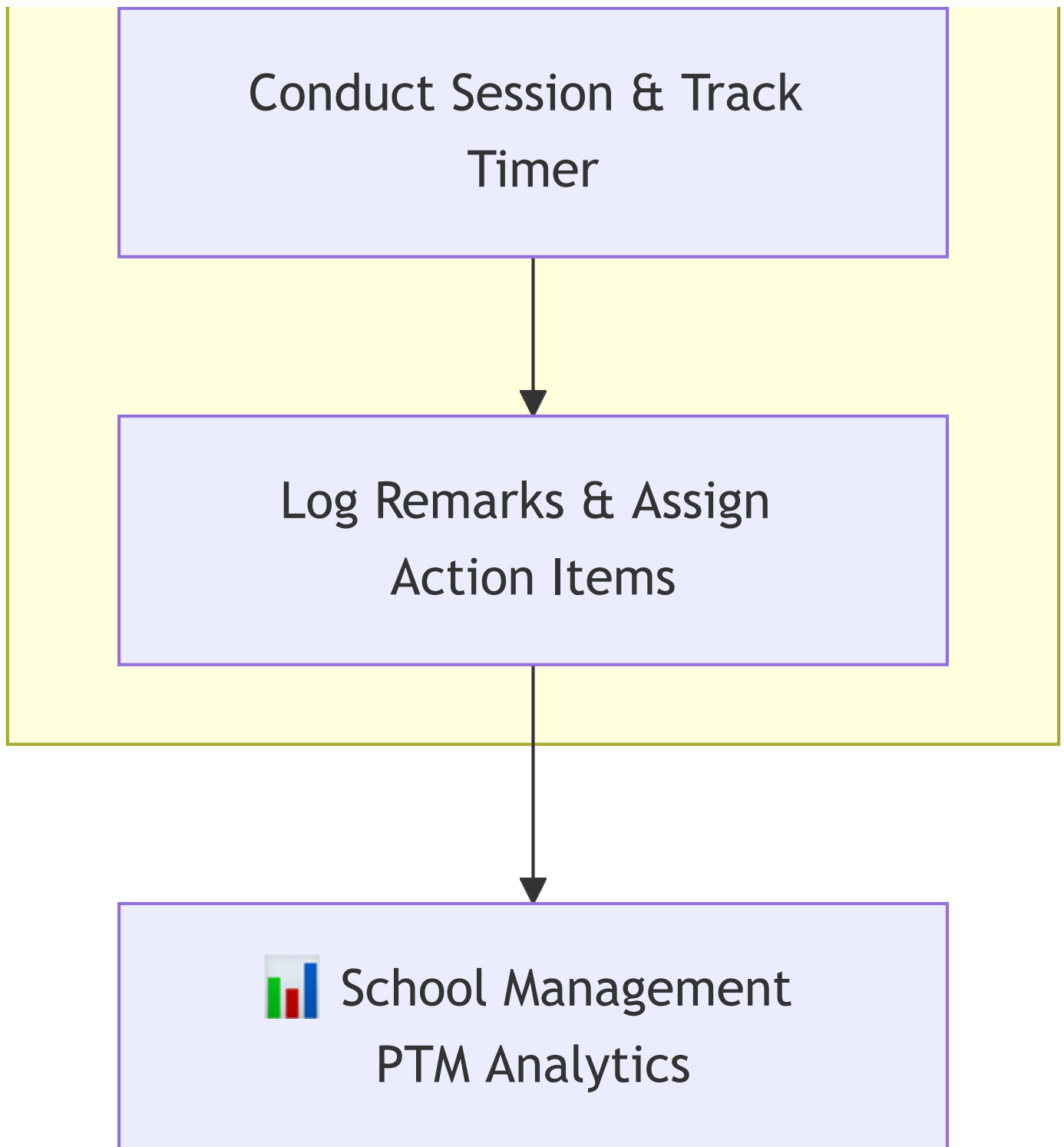


1. Module Workflow & System Architecture









2. Role-Based Feature Specifications

A. School Administrator Features

- **Event Configurator:**
 - Event title (e.g., "Mid-Term Academic Review 2026").
 - Target academic year, term, and scope (All grades / Specific class levels / Sections).
 - Date, time window (e.g., 09:00 AM – 01:30 PM), and slot duration (e.g., 10 mins , 15 mins , 20 mins).
 - Automated break times (e.g., 5-min gap between slots, 30-min tea break).

- **Meeting Modes:**
 - **In-Person:** Automated room and desk allocation (e.g., *Class 10-A, Room 102, Desk 1*).
 - **Virtual / Online:** Automated video link generation (Google Meet / Jitsi / Zoom integration).
 - **Hybrid:** Parents choose between in-person or online attendance.
- **Admin Control & Analytics Dashboard:**
 - Real-time slot booking percentage by class and teacher.
 - Overall parent participation rate (Booked vs Conducted vs No-Show).
 - School-wide PTM CSAT / Parent Feedback score.
 - Action item resolution tracking.

B. Parent Portal & Mobile App Features

- **Smart Interactive Slot Selector:**
 - Visual time-grid displaying open, booked, and unavailable slots for their child's class teacher and subject teachers.
 - **Multi-Child Intelligent De-confliction:** Auto-suggests non-overlapping back-to-back slots if a parent has 2+ children in different classes.
- **Pre-Meeting Agenda Builder:**
 - Parents can input specific queries or concerns prior to the session (e.g., *"Discussion regarding Mathematics term exam score"* or *"Bus pickup timing concern"*).
- **Live Queue & Navigation (For In-Person PTMs):**
 - Live session indicator (e.g., *"Currently meeting: Token #5 — Estimated wait time: 12 mins"*).
 - Room map and navigation guidance within the school premises.
- **Post-PTM Record:**
 - Access to digital PTM summary, teacher remarks, and agreed action items anytime via the app.

C. Teacher Portal Features

- **360° Student Performance Dossier:**
 - Selecting any booked slot immediately opens a comprehensive student summary card:
 - **Attendance:** Overall attendance %, recent absent days, punctuality record.
 - **Academics:** Recent exam scores, subject rank, progress graph over terms.
 - **Assignments:** On-time submission rate, pending assignments.
 - **Behavioral & Behavioral Flags:** Previous PTM notes, discipline logs, extracurricular activities.
- **Live Session Workspace:**
 - Interactive timer (e.g., 15-min countdown with visual warning at 2 mins remaining).
 - One-click extend session (+3 mins) or mark completed / no-show.
 - Quick feedback tag selection (Punctual , Attentive , Needs Practice in Math , Active in Class Discussions).
- **Action Item Manager:**
 - Create assignable action items with due dates (e.g., *"Assign remedial classes"*, *"Parent to supervise daily reading for 20 mins"*).

D. AI-COS Value Additions (AI Features)

1. AI Student Briefing Generator:

- Generates a 3-bullet talking point summary for teachers prior to each parent stepping in:
 - "1. Science performance increased by 18%."
 - "2. Missed 3 Hindi homework submissions in August."
 - "3. Parent previously requested update on football team selection."

2. AI Notes Refinement:

- Converts quick informal teacher notes (e.g., *"child distracted in class, needs multiplication practice, parent will monitor at home"*) into formal, well-structured feedback for the parent report.

3. Database Schema Design (Django ORM Models)

```
import uuid
from django.db import models
from tenants.models import TenantAwareModel
from academics.models import AcademicYear, ClassLevel, Section
from django.conf import settings

class PTMEvent(TenantAwareModel):
    """Overall PTM Event scheduled by School Admin"""
    MODE_CHOICES = [
        ('IN_PERSON', 'In Person'),
        ('VIRTUAL', 'Virtual'),
        ('HYBRID', 'Hybrid'),
    ]

    id = models.UUIDField(primary_key=True, default=uuid.uuid4, editable=False)
    title = models.CharField(max_length=200)
    academic_year = models.ForeignKey(AcademicYear, on_delete=models.CASCADE, related_name='ptm_events')
    date = models.DateField()
    start_time = models.TimeField()
    end_time = models.TimeField()
    slot_duration_minutes = models.PositiveIntegerField(default=15)
    gap_duration_minutes = models.PositiveIntegerField(default=5)
    break_start_time = models.TimeField(null=True, blank=True)
    break_end_time = models.TimeField(null=True, blank=True)
    mode = models.CharField(max_length=20, choices=MODE_CHOICES, default='IN_PERSON')
    target_class_levels = models.ManyToManyField(ClassLevel, related_name='ptm_events')
    is_published = models.BooleanField(default=False)
    created_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return f"{self.title} ({self.date})"

class PTMSlot(TenantAwareModel):
    """Individual time slots per teacher and section"""
    id = models.UUIDField(primary_key=True, default=uuid.uuid4, editable=False)
    event = models.ForeignKey(PTMEvent, on_delete=models.CASCADE, related_name='slots')
    teacher = models.ForeignKey('profiles.TeacherProfile', on_delete=models.CASCADE, related_name='ptm_s')
    section = models.ForeignKey(Section, on_delete=models.CASCADE, related_name='ptm_slots')
    start_time = models.TimeField()
    end_time = models.TimeField()
    room_number = models.CharField(max_length=50, blank=True, null=True)
    meeting_link = models.URLField(blank=True, null=True)
    is_booked = models.BooleanField(default=False)

    class Meta:
        ordering = ['start_time']

    def __str__(self):
        return f"{self.teacher.user.last_name} | {self.section.name} | {self.start_time}-{self.end_time}"
```

```
class PTMBooking(TenantAwareModel):
    """Parent booking and session recording"""
    STATUS_CHOICES = [
        ('BOOKED', 'Booked'),
        ('IN_PROGRESS', 'In Progress'),
        ('COMPLETED', 'Completed'),
        ('NO_SHOW', 'No Show'),
        ('CANCELLED', 'Cancelled'),
    ]

    id = models.UUIDField(primary_key=True, default=uuid.uuid4, editable=False)
    slot = models.OneToOneField(PTMSlot, on_delete=models.CASCADE, related_name='booking')
    student = models.ForeignKey('profiles.StudentProfile', on_delete=models.CASCADE, related_name='ptm_b')
    parent = models.ForeignKey('profiles.ParentProfile', on_delete=models.CASCADE, related_name='ptm_boo')

    parent_pre_query = models.TextField(blank=True, null=True, help_text="Pre-meeting query submitted by status")
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='BOOKED')

    teacher_remarks = models.TextField(blank=True, null=True, help_text="Shared feedback visible to pare")
    teacher_private_notes = models.TextField(blank=True, null=True, help_text="Internal staff notes")
    parent_rating = models.PositiveIntegerField(null=True, blank=True, help_text="1 to 5 Star Parent Sat")

    started_at = models.DateTimeField(null=True, blank=True)
    completed_at = models.DateTimeField(null=True, blank=True)

    def __str__(self):
        return f"Booking: {self.student.user.first_name} with {self.slot.teacher.user.last_name}"

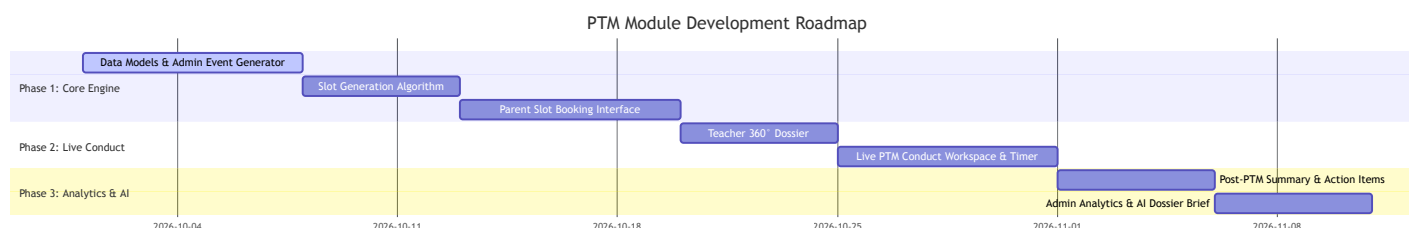
class PTMActionItem(TenantAwareModel):
    """Action items generated during a PTM session"""
    ASSIGNED_TO_CHOICES = [
        ('PARENT', 'Parent'),
        ('STUDENT', 'Student'),
        ('TEACHER', 'Teacher'),
        ('COUNSELOR', 'Counselor'),
    ]

    id = models.UUIDField(primary_key=True, default=uuid.uuid4, editable=False)
    booking = models.ForeignKey(PTMBooking, on_delete=models.CASCADE, related_name='action_items')
    description = models.CharField(max_length=255)
    assigned_to = models.CharField(max_length=20, choices=ASSIGNED_TO_CHOICES)
    due_date = models.DateField(null=True, blank=True)
    is_completed = models.BooleanField(default=False)
    completed_at = models.DateTimeField(null=True, blank=True)

    def __str__(self):
        return f"Action Item: {self.description} ({self.assigned_to})"
```



4. Phased Rollout & Roadmap





5. Key Metrics for Success (KPIs)

- **Queue Elimination Rate:** 90%+ reduction in physical waiting lines at school.
- **Parent Slot Utilization:** >85% of parents booking their slots digitally.
- **Teacher Preparation Time:** Reduced from ~10 mins per student to instant (<10 seconds) via the 360° Dossier.
- **Parent Satisfaction Score (CSAT):** Target > 4.5 / 5.0.